Rick den Otter

Apeldoorn, the Netherlands   rickdotyt@gmail.com
0009-0002-8852-1697   rick-denotter   rickdott

Education

PhD **Experimental Psychology, Utrecht University**, Artificial Intelligence & Cognitive Neuroscience – Utrecht, the Netherlands — May 2024 – present
My PhD is a continuation of my masters' thesis project, where I apply Hidden Multivariate Pattern analysis combined with deep spatiotemporal sequence models to decode cognitive operations from EEG at the trial level.

MSc **Utrecht University**, Artificial Intelligence – Utrecht, the Netherlands — Feb 2022 – Feb 2024
I conducted my thesis at the department of Experimental Psychology (Utrecht University) on classifying cognitive processing operations from EEG using machine learning classifiers. During this thesis I was chosen to present my research at the NVP Dutch Society for Brain and Cognition Winter Conference. My thesis was awarded a grade of 8.9, with a 9.5 for both the result and the process.

BSc **Hogeschool van Amsterdam**, HBO-ICT Software Engineering – Amsterdam, the Netherlands — Sept 2017 – June 2021
- Graduated Cum Laude (Grade: 8.3/10)
- Followed thematic semester "Big Data" and minor "Positive Psychology & Technology"
- Best first year project out of 150+ groups as voted by peers and teachers (2017)

HAVO High School, General Education – Lelystad, the Netherlands — 2016
Profile Nature & Health

Publications

Trial-level sequence modeling reveals hidden dynamics of dual-task interference — May 2026
Rick den Otter, Anna Dame, Sjoerd Stuit, Leendert van Maanen
10.1371/journal.pcbi.1014302 (PLOS Computational Biology)

Identifying trial-by-trial cognitive strategies in speed-accuracy trade-off through deep sequence modeling — under review
Rick den Otter, Gabriel Weindel, Sjoerd Stuit, Leendert van Maanen

A Trial-Level Neural Predictor of Proactive Slowing and Stop-Signal Success — in preparation
Rick den Otter, Leendert van Maanen, Leon Kenemans, Daan van Rooij

Grants and Awards

NWO Small Compute Grant (310.000 compute units on Snellius) — Feb 2026
Rick den Otter

Conferences

NVP Dutch Society for Brain and Cognition Winter Conference — Dec 2025
Egmond aan Zee, the Netherlands (Poster presentation)

International Conference on Cognitive Neuroscience (ICON) — Sept 2025
Porto, Portugal (Poster presentation)

Cognitive Computational Neuroscience (CCN) July 2025
Amsterdam, the Netherlands (Poster presentation)

Helmholtz Institute - Perception day Nov 2024
Utrecht, the Netherlands (Poster presentation)

NVP Dutch Society for Brain and Cognition Winter Conference Dec 2023
Egmond aan Zee, the Netherlands (Poster presentation)

Academic Service

(Co)-reviewer: eLife, Cognitive Computational Neuroscience conference

Teaching Experience

Machine Learning for Human Vision & Language (MSc Artificial Intelligence, Utrecht University): Teaching work groups, grading exams

Human Network Analysis (MSc Applied Data Science, Utrecht University): Teaching work groups, grading exams

Programming (BSc Psychology, Utrecht University): Guiding and grading students during their programming assignments

Thesis supervision (BSc Artificial Intelligence, Utrecht University): Guiding and grading students during their BSc thesis

Thesis supervision (MSc Artificial Intelligence, Utrecht University): Guiding and grading students during their MSc thesis

Professional Experience

Software Engineer, Thesis Intern, Intern, Sweco Nederland – De Bilt, the Netherlands Jan 2019 – Feb 2024
My internships and job at Sweco Nederland started with developing and maintaining the REST API for Obsurv. Later, I developed a novel application handling concurrent long-running export tasks from a relational database to RDF triple graph format. Finally, I conducted my BSc thesis at Sweco Nederland on using machine learning to recognize defects in sewer inspection media. I integrated this model into a deployed product using Azure Machine Learning. Additionally, I guided multiple interns, assisting them with the technical part of their internships, and grading their theses. Technologies used: PL/SQL, Oracle stack, TypeScript, JavaScript, Express, NodeJS, Python, VR in Three.JS, Azure for DevOps activities, Azure Machine Learning.

Courses and Summer Schools

Smoothening your writing process July 2025
Utrecht, the Netherlands

Inclusion and Social Safety training Dec 2024
Utrecht, the Netherlands

Summer School Advanced Communication Skills Aug 2024
Utrecht, the Netherlands

Skills

Programming: Proficient: Python (PyTorch, Pandas, NumPy), R (lme4), MATLAB (EEGLAB). Familiar: C#, JavaScript, HTML/CSS

Other technical skills: Git/GitHub, Linux, Docker, Distributed computing (Azure, Snellius), Adobe Creative Cloud (Photoshop, Illustrator, InDesign)

Experimental methods: EEG (BioSemi ActiveTwo)

Languages: English (fluent), Dutch (native)